

Simple Direct Estimation of Link SMD and MDL for MIMO-DSP Design in Ultra-Wideband Coupled-Core Fiber Transmission

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Abstract We propose a simple optical setup for estimating SMD and MDL in coupled-core fibers over a 150-nm bandwidth without full MIMO-DSP characterization. The method is validated by measurements on 4- and 12-core fibers and by C+L-band WDM MIMO transmission over a field-installed 12-core link. ©2026 The Author(s)

Introduction

Coupled-core multi-core fiber (CCF)-based space-division multiplexing (SDM) [1] is a promising approach for scaling transmission capacity while maintaining a standard cladding diameter [2]. Extending CCF transmission to multiband wavelength-division multiplexing (WDM) beyond the C and L bands is therefore an important step toward further capacity scaling [3] [4]. For practical system deployment, key link parameters such as spatial-mode dispersion (SMD) and mode-dependent loss (MDL), which depend on the specific details of fiber connections and bending conditions in the installed link and affect DSP parameters such as the required equalizer tap length and feasible modulation order, should be characterized in advance.

In conventional transmission experiments, however, the most common approach has been to estimate the SMD and MDL only after full signal transmission and demodulation, based on the MIMO equalizer taps in the DSP [5] [6]. This approach involves several challenges. First, it is difficult to prepare a highly complex MIMO receiver, especially in less conventional wavelength bands such as the S and U bands. Second, with respect to MDL, it is impossible to separate the MDL originating from the Tx/Rx subsystem from the link MDL, which leads to overestimation of the link MDL, particularly in cases such as single-span transmission. Optical vector network analyzers [7] alleviate some of these challenges but still suffer from limited scalability with respect to the number of modes and sensitivity to vibration. Although measurement techniques have been developed in the fiber-design community for evaluating SMD and MDL [8] [9], respectively, the metrics used in fiber design and those used in DSP are not directly equivalent [10], and no comprehensive, wavelength-resolved experimental comparison has been reported.

The contributions of this work are as follows.

- We propose simple optical configurations for evaluating the SMD and MDL of

CCF without full MIMO-DSP. The method is scalable with respect to the number of modes and the operating WDM band. To the best of our knowledge, this is the first method that enables measurement of SMD and MDL using common hardware configuration with these features.

- We evaluate 4-core CCF (4CCF) in the laboratory and 12-core CCF (12CCF) in both laboratory and field environments over a 150-nm bandwidth spanning the S to U bands. For a field-installed 12CCF, we compare the measured values with DSP-based results obtained from a C+L-band WDM MIMO transmission setup.

These results provide a basis for establishing design guidelines for wideband WDM CCF systems.

Principle of proposed method

Figure 1 illustrates the proposed method, where both parameters are estimated by switching the measurement condition of a single optical setup. Unlike conventional methods for MDL/SMD characterization such as MIMO-DSP tap analysis and optical vector network analyzer, which do full characterization of the complex-valued transmission matrix, the proposed method does not require simultaneous measurement of all input-output tributaries. This reduces measurement complexity, especially for high-core-count fibers. In the setup, a frequency-swept laser is launched into one channel through a fan-in device. After passing through the mechanical scrambler (MS), the fiber under test (FUT), and the knife-edge beam block, the optical signal is detected by a photodiode (PD) with a sufficiently large active

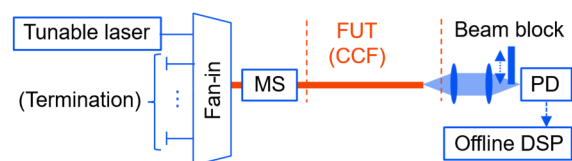


Fig. 1: Proposed setup. MS: mechanical scrambler, FUT: fiber under test, PD: photodetector.

area to neglect the spatial profile. SMD and MDL are evaluated by changing the states of the MS and the beam block. In SMD measurement, the measurement can be regarded as an extension of the fixed-analyzer (FA) method [8]. In this case, the MS is stopped, and the knife-edge beam block is adjusted so that part of the incident optical power is blocked (in this case, the block is placed at the image plane of the fiber facet). Let the number of modes, including both spatial and polarization degrees of freedom, be denoted by D , and let the $D \times D$ transmission matrix be $H(f)$. The (i, j) element is denoted by $h_{ij}(f)$. Then, the detected intensity $I(f)$ can be approximated as follows:

$$I(f) \propto \sum_{i=1}^D a_i |h_{1i}(f)|^2, \quad a_i > 0, \quad (1)$$

and a_i are not all equal. Here, we make use of the fact that the fiber under test is a CCF. Specifically, since the distributions of the individual cores are separated at the imaging plane, the inter-core interference terms can be neglected. In CCF, the autocorrelation function satisfies

$$R_I(\Delta f) = \langle I(f)I(f + \Delta f) \rangle \propto \exp\left(-\frac{\Delta f^2}{2\sigma_f^2}\right), \quad (2)$$

where σ_f is the standard deviation of the Gaussian profile. The temporal spread parameter σ is then expressed as $\sigma = 1/(\sqrt{2}\pi\sigma_f)$. In the fiber-design community, the SMD of a CCF is commonly expressed in units of ps/km^{1/2}. For a fiber of length L , it is written as $\text{SMD}_{\text{fiber}} = \sigma/\sqrt{L}$.

An important point is the conversion between the SMD used in the fiber-design community, i.e., $\text{SMD}_{\text{fiber}}$ [ps/km^{1/2}], and the SMD metric τ [ps] often used in transmission-system design. The latter is frequently defined as the temporal width containing such as 90% or 99% of the energy of the intensity impulse response. These two quantities can be related by

$$\tau = 2C_{\text{SMD}}\text{SMD}_{\text{fiber}}\sqrt{L}, \quad (3)$$

where C_{SMD} is a coefficient determined such that

$$E = \int_{-C_{\text{SMD}}}^{C_{\text{SMD}}} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right) dx \quad (4)$$

where E is an arbitrary energy threshold such as 90% (0.9). The expressions for C_{SMD} for some E are summarized in Table 1.

MDL can also be evaluated with the same setup by turning on the MS and removing the beam block ($I(f) \propto \sum_{i=1}^D |h_{1i}(f)|^2$). This condition

Tab. 1: C_{SMD} with respect to E .

E	C_{SMD}
90 %	1.645
95 %	1.960
99 %	2.576
99.9 %	3.291

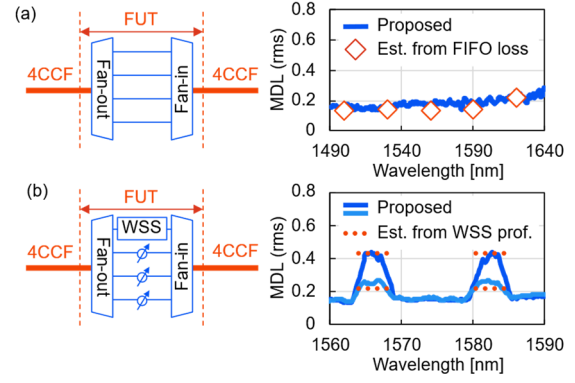


Fig. 2: Example results of wavelength-resolved MDL measurements. (a) FIFO pair. (b) FIFO pair with WSS.

is equivalent to the scrambling method [9]. Under the assumption that the distribution of singular value of $H(f)$ is Gaussian, MDL in is

$$\text{MDL}_{\text{rms}} = \frac{10}{\log 10} \sqrt{\log \left(1 + \frac{(D+1)\text{Var}(I(f))}{\langle I(f) \rangle} \right)} \quad (5)$$

which is defined as the rms variation of the singular values of $H(f)$. Note that this definition differs from that used in Ref. [9] by a factor of approximately $\sqrt{D+1}$ [11]. In the present study, the MS operates at a rate of several Hz, consistent with the previous work [9].

Figure 2 shows the results of experiments conducted to verify the feasibility of wavelength-resolved MDL estimation with a single sweep. A four-core fiber was used as the FUT. In Fig. 2(a), a pair of Fan-in/Fan-out (FIFO) was inserted as an MDL source. The wavelength was swept at 20 nm/s. The measured MDL, shown by the blue curve, agrees well with the value calculated from the loss of the FIFO devices. In Fig. 2(b), a wavelength-selective switch (WSS) was inserted into one channel of the FIFO pair, while variable optical attenuators were inserted into the other channels to equalize the baseline loss. The WSS was configured to introduce an additional loss of 0.5 dB or 1 dB over the wavelength ranges 1563.96-1568.05 nm and 1580.45-1584.63 nm, and 0 dB elsewhere. The measured results reproduce well the expected MDL_{rms} values indicated by the dashed lines, confirming that a spectral resolution of several nms is achievable.

Results

Figure 3 shows setups for measurements of CCFs for transmission link. Figure 3(a) shows the experimental setup. The fiber under test was a 53-km 4CCF [10], 51-km 12CCF [12], and total 68-km (4.86×14-km) field-installed 12CCF [13]. MIMO transmission was also performed using a C+L-band WDM setup for the field-installed 12CCF. The MIMO transmission setup was almost identical to that reported in [13][14], except

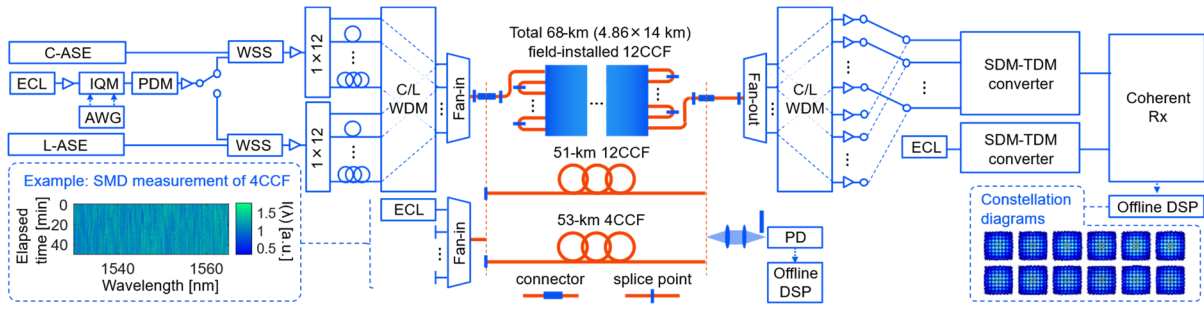


Fig. 3: Experimental setup. ASE: amplified spontaneous emission, IQM: in-phase/quadrature modulator, AWG: arbitrary wave-form generator, PDM: polarization division multiplexing, ECL: external cavity laser, TDM: time-division multiplexing.

that the symbol format was probabilistically shaped QAM at 144 Gbaud and an additional L-band setup including a C/L WDM coupler was employed. The total WDM bandwidth was 10.2 THz, covering 1528.58-1565.70 nm and 1573.09-1620.28 nm. In the proposed measurement configuration, a wavelength-swept lasers covering 1490-1640 nm was used. In the inset, as an example, we show the results of repeated measurements of $I(\lambda)$ in the SMD estimation of 4CCF, arranged in chronological order. Note that, although this is beyond the scope of the present work, this method may make it possible to evaluate the timescale of temporal fluctuations of $H(\lambda)$.

Figure 4 shows the SMD measurement results. Here, measured τ values with $E = 0.9$ and are displayed with 50-time averaging. In the measurement of the field-installed 12CCF, the proposed method and the MIMO-DSP-based method showed good agreement, with an averaged difference of 11 ps. This corresponds to a length of less than two symbols in this setup. In all fibers, the SMD showed wavelength dependence; however, the dependence can be controlled through the fiber or cable design [15][16].

Figure 5 shows the MDL measurement results. Fig. 5(a) shows a comparison among the three types of fiber. Only the lab 12-core fiber exhibited clear wavelength dependence, which may be attributed to bending-loss characteristics. The MDL values estimated from the MIMO tap coefficients were approximately 1 dB higher than those obtained by the proposed configuration, due to power imbalance in the input components in the Tx/Rx subsystem, such as the WDM coupler. This result demonstrates that the proposed configuration can isolate the fiber contribution and

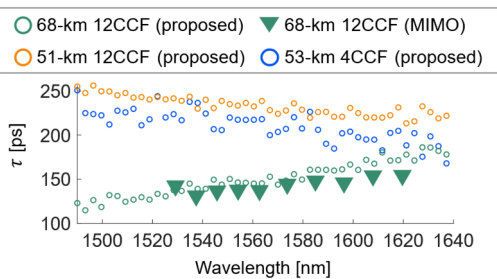


Fig. 4: Results of SMD measurements of link CCFs.

evaluate the MDL of the fiber itself.

In addition, the field-installed 12CCF showed clearly higher MDL than the lab fiber, with an average of 0.31 dB, because this fiber was constructed by connecting 14 fiber sections using fusion splices and connectors [17]. Fig. 5(b) shows the distance dependence. The increase in MDL with distance is observed, demonstrating the applicability of the proposed method to the evaluation of future field-deployed systems. The present MDL/SMD measurement results also support the feasibility of ultra-wideband transmission over more than 150 nm in 4-core and 12-core CCFs.

Conclusion

We demonstrated a simple MIMO-less method for evaluating SMD and MDL in CCF using a common optical setup. The SMD results showed good agreement with MIMO-DSP-based estimates, whereas the MDL comparison clarified that DSP-based values include subsystem imbalance in addition to the fiber contribution. These results indicate that the proposed method is effective for link-specific impairment characterization and provides a practical basis for pre-design of MIMO-DSP parameters in future wideband SDM/WDM transmission systems.

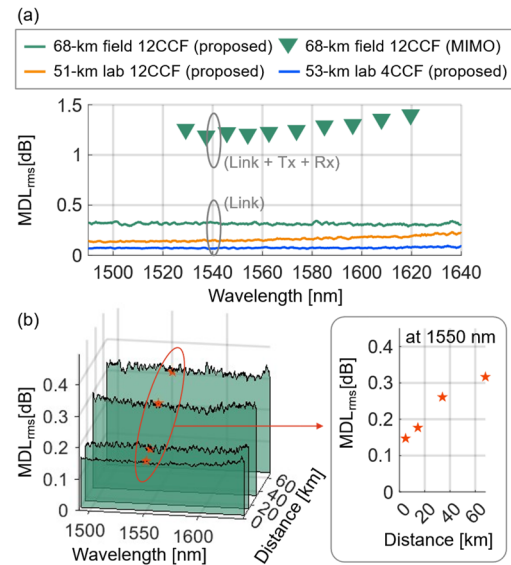


Fig. 5: (a) Results of MDL measurements of link CCFs. (b) Distance dependence of MDL of field-installed 12CCF.

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